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Digital product passport - Unique identifiers

Passeport numérique des produits - Identifiants uniques

Digitaler Produktpass - Eindeutige Kennungen

This draft European Standard is submitted to CEN members for enquiry. It has been drawn up by the Technical Committee CEN/CLC/JTC 24.

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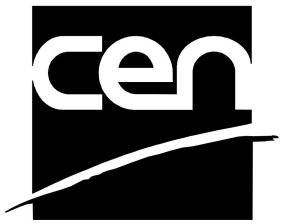

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European foreword

This document (prEN 18219:2025) has been prepared by Technical Committee CEN/CLC/JTC 24 "Digital product passport – Framework and system", the secretariat of which is held by DIN.

This document is currently submitted to the CEN Enquiry.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

This document has been prepared as one of eight standards developed under Standardization Request M/604, issued by the European Commission to CEN and CENELEC.

Introduction

0.1 General

In 2019, the European Commission introduced the European Green Deal, a detailed strategic plan aimed at sustainable growth. A key component of this plan is the Ecodesign for Sustainable Product Regulation (ESPR) [1], which is designed to guide the European Union towards achieving climate neutrality by 2050. The ESPR promotes a vision of a society that is equitable and thriving, functioning within a modern, competitive, and circular economy, all while preserving a toxin-free environment.

In detail, the ESPR mandates the adoption of production and consumption patterns that are in harmony with the Union's comprehensive sustainability goals related to climate change, environmental protection, energy use, resource efficiency, and biodiversity conservation, all within defined planetary boundaries. To achieve these goals, the regulation introduces a set of stringent ecodesign requirements. These are specifically crafted to enhance the durability, reliability, repairability, upgradability, reusability, and recyclability of products. Such measures are vital for reducing waste, diminishing the presence of hazardous substances in products, and improving their energy and resource efficiency. Collectively, these requirements establish a rigorous framework for sustainability within the industry, pivotal for supporting the Union's transition to sustainability.

To support the implementation of these comprehensive requirements, the ESPR highlights Digital Product Passports (DPP) as a key element and a pivotal tool to ensures that all stakeholders in the value chain – manufacturers, distributors, consumers, and recyclers – have access to essential, traceable, and reliable product information, supporting informed consumer choices, and promoting better resource management and sustainability practices.

The rollout of the Digital Product Passport (DPP) concept is set to occur gradually over the coming years. Specific details, including what will be included in the DPP for various product groups will be outlined in delegated acts. This approach ensures a comprehensive strategy that balances environmental objectives with regulatory requirements and stakeholder expectations. This also includes other EU policies and regulations, such as construction products or batteries.

To support the implementation of the DPPs, this document establishes a framework for the creation and management of unique identifiers. These identifiers connect products, economic operators, and facilities to their digital counterparts, and are designed to be distinct, reliable, and universally recognizable. They are crucial for establishing robust and resilient digital product passports, enabling effective tracking of environmental attributes and compliance across the life cycle. This improves the accessibility of product information through data carrier scans, giving consumers, business partners, and regulators access to a digital product passport. This digital passport provides valuable insights into the product's durability, reparability, recycled content, availability of spare parts, and more.

To ensure interoperability and reduce implementation costs for businesses, these identifiers are designed to integrate with existing International Standards and align with current technological advancements, while maintaining consistency and coherence with established industry practices.

0.2 Orientation

This document provides the foundation for the identification of products, economic operators, and facilities to be used for digital product passports.

Clause 4 sets out the principles, requirements, and guidelines that apply across all identification use cases. These are structured around six key areas: global uniqueness, persistence, syntax, semantics, interoperability, and openness.

— *Principles:* Provide the intent and purpose for each area.

— *Requirements:* Define what must be fulfilled when implementing identifiers and identification schemes, serving as the foundation for assessment and compliance.

— *Guidelines:* Offer optional recommendations that can support or enhance implementation but are not mandatory.

Clause 5 specifies the permitted identification schemes for creating a unique product identifier. Five different schemes are described, each of which may be used to meet the requirements established in Clause 4, provided that both the rules of the chosen scheme and the requirements of Clause 4 are fully satisfied.

Clause 6 specifies the permitted identification schemes for creating a unique economic operator identifier and unique facility identifier. Four schemes are described, each of which may be used to meet the requirements established in Clause 4, provided that both the rules of the chosen scheme and the requirements of Clause 4 are fully satisfied.

In addition, the document includes informative annexes to support practical understanding and implementation:

— *Annex A*: Guidance on selecting item-level identification for products.

— *Annex B*: Overview of identification schemes for products.

— *Annex C*: Overview of identification schemes for economic operators and facilities.

— *Annex ZA:* Relationship between this European Standard and the essential requirements of 2024/1781.

1 Scope

This document defines the principles and specifies the requirements and guidelines for unique product identifiers, unique economic operator identifiers, and unique facility identifiers used in digital product passports. It covers the following areas:

a) global uniqueness;

b) persistence;

c) syntax;

d) semantics;

e) interoperability;

f) openness.

This document accommodates unique product identifiers at three granularity levels of specificity: model, batch, or individual item, to support various operational needs.

This document describes identification (ID) schemes that use issuing agencies, self-issuing systems, or a combination of both.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 646:1991, *Information technology -- ISO 7-bit coded character set for information interchange*

ISO/IEC 6523-1:2023, *Information technology — Structure for the identification of organizations and organization parts — Part 1: Identification of organization identification schemes*

ISO/IEC 6523-2:1998, *Information technology -- Structure for the identification of organizations and organization parts -- Part 2: Registration of organization identification schemes*

ISO/IEC 15418:2016, *Information technology - Automatic identification and data capture techniques - GS1 Application Identifiers and ASC MH10 Data Identifiers and maintenance*

ISO/IEC 15434:2019, *Information technology — Automatic identification and data capture techniques — Syntax for high-capacity ADC media*

ISO/IEC 15459-2:2015, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 2: Registration procedures*

ISO/IEC 15459-4:2014, *Information technology -- Automatic identification and data capture techniques -- Unique identification -- Part 4: Individual products and product packages*

ISO/IEC 15459-6:2014, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 6: Groupings*

ISO/IEC 15961-1:2021, *Information technology - Data protocol for radio frequency identification (RFID) for item management - Part 1: Application interface*

ISO/IEC 17360:2023, *Automatic identification and data capture techniques — Supply chain applications of RFID — Product tagging, product packaging, transport units, returnable transport units and returnable packaging items*

ISO 17442-1:2020, *Financial services — Legal entity identifier (LEI) — Part 1: Assignment*

ISO/IEC 18975:2024, *Information technology — Automatic identification and data capture techniques — Encoding and resolving identifiers over HTTP*

ISO/IEC 19762:2016, *Information technology — Automatic identification and data capture (AIDC) techniques — Harmonized vocabulary*

ISO 26324:2025, *Information and documentation — Digital object identifier system*

EN IEC 61406-1:2022, *Identification Link - Part 1: General requirements*

EN IEC 61406-2:2024, *Identification link - Part 2: Types/models, lots/batches, items and characteristics*

Decentralized Identifier Resolution (DID Resolution) v0.3:2025, *https://www.w3.org/TR/did-resolution/*

Decentralized Identifiers (DIDs) v1.0:2022, *https://www.w3.org/TR/did-1.0/*

Verifiable Credentials Data Model v1.1:2022, *https://www.w3.org/TR/vc-data-model/*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 19762:2016 and the following apply.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
batch
subset of a model that is grouped by the economic operator based on the identical properties

3.2
consumer
individual member of the general public purchasing or using goods, property or services for private purposes

[SOURCE: ISO 14025:2006]

3.3
digital product passport
DPP
digital record of product characteristics throughout its life cycle

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability.

3.4
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider

3.5
identification scheme
ID scheme
system for allocating identifiers to registered objects

[SOURCE: ISO/IEC 6523-1:2023]

3.6
interoperability
ability of independent systems to exchange meaningful information and initiate actions from each other, in order to operate together to mutual benefit

[SOURCE: ISO/IEC 26324:2025, 3.11]

3.7
issuing agency
centralized identification scheme
federated identification scheme
organization that allocates company codes, defines rules and/or identification schemes

3.8
item
single unit of a model

3.9
life cycle
consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal

[SOURCE: ISO 14040:2006]

3.10
model
version of a product of which all units share the same technical characteristics and the same model identifier

3.11
object
entity to which specified requirements apply

Note 1 to entry: Product, process, service, system, installation, project, data, design, material, claim, person, body or organization, or any combination thereof.

[SOURCE: ISO/IEC 17000:2020]

3.12
persistence
existence, and ability to be used in services outside the direct control of the issuing assigner, without a stated time limit

[SOURCE: ISO 26324:2022]

3.13
placed on the market
first making available of a product on the market

3.14
product
any physical good that is placed on the market or put into service

3.15
put into service
first use, for its intended purpose, of a product

3.16
resolution
process in which an identifier translates or maps to associated information or metadata about the object it represents

Note 1 to entry: DNS resolution and HTTP/HTTPS resolution utilize standard web functionality and do not require additional software. In contrast, DID resolution typically requires additional software.

3.17
resolver
service that accepts an identifier or parameters as input and responds in some way specific to that input

Note 1 to entry: An economic operator can set up a resolver by themself or use a service of a designated service provider.

3.18
self-issuing system
decentralized identification scheme
system or mechanism that an organization uses to generate and assign unique identifiers to its objects without the intervention or oversight of an external authority

3.19
unique economic operator identifier
unique string of characters for the identification of actors involved in a product's value chain

3.20
unique facility identifier
unique string of characters for the identification of locations or buildings involved in a product's value chain or used by actors involved in a product's value chain

3.21
unique identifier
identifier which is guaranteed to be unique among all identifiers used for those objects and for a specific purpose

Note 1 to entry: A unique identifier refers to *unique product identifier* (3.22), *unique economic operator identifier* (3.19), and *unique facility identifier* (3.20).

[SOURCE: ISO 29404:2015, modified – Note 1 to entry added]

3.22
unique product identifier
unique string of characters for the identification of a product that also enables a web link to the digital product passport

4 General principles, requirements and guidelines

4.1 Global uniqueness

4.1.1 Principle

The principle of global uniqueness ensures that each identifier is distinct, at least exclusive within its domain of use, and non-coexisting, enabling clear object differentiation and preventing misidentification. Products (3.14) receive at least one globally unique product identifier (3.22) at one of the three levels of granularity (model, batch, or individual item), and when necessary, economic operator or facility is assigned a globally unique economic operator identifier (3.19) or a globally unique facility identifier (3.20). The economic operator and/or issuing agency ensures this uniqueness when placing products on the market (3.13) or putting them into service (3.15).

4.1.2 Requirements

The unique identifier shall be globally unique, by ensuring:

1) *No duplication:* The same unique identifier shall not be used to identify different objects.

2) *Distinct:* The unique identifier shall be unique at least within its domain and across all potential contexts in which it may be used.

3) *Non-coexistence:* A unique identifier, once issued, shall not coexist with or be reassigned to another object.

4.2 Persistence

4.2.1 Principle

The principle of persistence (3.12) ensures that the unique identifier remains associated with the same object at least throughout its life cycle (3.9). The purpose of a persistent identifier is to accurately represent an object for as long as it exists, regardless of changes in company status, ownership, location, including mergers, acquisitions, splits or spin-offs. The economic operator is responsible for recording, maintaining, and securely storing each identifier, either directly or through a service provider, to ensure persistency.

4.2.2 Requirements

The unique identifier shall provide persistence, with the economic operator responsible for:

1) *Consistency:* The unique identifier, once assigned, shall remain unchanged and consistently refer to the same object without ambiguity, for as long as it remains within the control of the economic operator.

2) *Preservation:* The unique identifier shall be preserved, at least throughout its expected life cycle (3.9), or as long as required by applicable regulations.

3) *Permanence:* The unique identifier shall remain available including after an insolvency, a liquidation, or a cessation of the economic operator responsible for its creation.

NOTE For products placed on the European market, the EU DPP registry ensures a minimum level of preservation and permanence of identifiers.

4.3 Syntax

4.3.1 Principle

The principle of syntax establishes a clear, standardized structure for unique identifiers, ensuring consistent recognition and usage across systems, services, and platforms when needed. This structure supports interoperability and efficient processing, by enabling the mapping between different syntaxes. The unique identifiers maintain a distinguishable format to enhance identification, searchability and retrieval. The length is specified for optimized processing and storage, and the syntax structure enables accessibility and usability on smartphones, other similar consumer devices and common web browsers.

4.3.2 Requirements

The unique identifier shall follow a predefined syntax, by ensuring:

1) *Character set:* The character set for the unique identifier shall include alphanumeric characters and, where necessary, percent encoding and specific symbols as set out in ISO/IEC 646:1991.

2) *Web:* The unique product identifier shall either be in the form of a URL, or have specified transformation into a URL (e.g. resolvable from URI to URL), ensuring compatibility with web-based systems.

4.3.3 Guidelines

1) *Structure:* The unique identifier may conform to a predefined structural pattern, ensuring it is easily parsed and recognized across various systems, services and platforms.

2) *Variable length:* Unique identifier may use variable length as long it does not exceed limitations of the chosen ID scheme.

4.4 Semantic

4.4.1 Principle

The principle of semantics ensures that a unique identifier accurately represents the object it refers to, when needed. Semantics also enable effective access to the data encoded within the identifier, supporting interpretation by both systems and users. Identifiers are aligned with the characteristics of the objects they represent, including distinctions between different types of identifiers and varying levels of granularity. Additionally, identifiers facilitate access to a digital product passport—either directly, via a data carrier, or through a resolution (3.16) mechanism.

4.4.2 Requirements

The unique identifier shall maintain specified semantics, by ensuring:

1) *Granularity:* Each product identifier shall be unique at least at the smallest granularity level it serves, whether it is a model (3.10), batch (3.1), or item (3.8).

2) *Granularity consistency:* The granularity level of a unique product identifier shall remain consistent once the product is placed on the market or put into service (3.15).

3) *Change in granularity:* If a change in granularity becomes necessary that has not been foreseen, a new or extended unique product identifier shall be assigned. The new or extended identifier shall be linked to the old unique identifier to maintain traceability.

4.4.3 Guidelines

1) *Granularity level:* The economic operator may adopt a more granular level of specificity for the unique product identifier than mandated by regulation (e.g., using batch instead of model, or item instead of batch) to enhance traceability or meet specific operational needs.

2) *Resolver:* The economic operator may utilize a resolver (3.17) for the resolution of a unique identifier, either established by themselves or provided by a designated service provider.

3) *Hierarchy:* A unique product identifier may be structured to represent multiple levels of granularity, such as model, batch and individual item identification, within a single hierarchical format.

4.5 Interoperability

4.5.1 Principle

The principle of interoperability (3.6) is enabled by the global uniqueness and the digital product system. In addition, unique identifiers ensure seamless operation across systems, services, and platforms without modification of the identifier or the platform, enabling consistent interpretation and unambiguous data exchange. A structured format supports compatibility with external components to enable automated processing and integration across industries. Portability ensures continuity across technological and organizational transitions. Key interoperability interfaces include data carriers, web links, APIs, and digital product passports.

4.5.2 Requirements

The unique identifier shall be interoperable, by ensuring:

1) *Data carrier:* The unique product identifier shall be retrievable from a data carrier specified in EN JTO24002, linking it to the digital product passport (3.3).

2) *Digital copy:* The economic operator shall give retailers and online marketplaces access to the unique product identifier that they need to lead consumers to the DPP of a given product.

3) *Compatibility:* The unique product identifier shall be compatible with external components, such as professional scanning devices, smartphones and similar digital consumer devices.

4) *Portability:* The unique identifier shall be portable across different systems, services, and platforms without losing their meaning or function.

5) *Existing ID:* Wherever possible, the economic operator shall be permitted to continue using their existing ID scheme to generate unique identifiers for the digital product passport.

4.6 Openness

4.6.1 Principle

The principle of openness promotes transparency by ensuring that policies, processes, and standards are publicly accessible, allowing stakeholders to study and implement them. The use of widely accessible rules and procedures foster broad adoption and interoperability. Identifiers are used to access DPP public information without requiring registration, download DPP specific software, or authentication. Openness aims to avoid unnecessary restrictions that could limit access, use, or innovation. Openness also facilitates the transition or switching between systems, services, and platforms without compromising functionality or data integrity.

4.6.2 Requirements

The unique identifier shall remain open and transparent, by ensuring:

1) *Transparency:* The ID scheme and unique identifier shall be based on a system where policies, processes, and standards can be studied, influenced, and used under fair, reasonable, and non-discriminatory terms.

2) *End consumer usage:* For end consumer (3.2) usage, the unique identifier shall be useable to access DPP public information without the need to register, download DPP-specific software, or add user credentials.

3) *No undue restrictions:* The unique identifier shall not result in vendor lock-in. It shall also not impose limitations on use, except where such restrictions are technically necessary.

4.6.3 Guidelines

1) *Smartphones and similar devices:* The ID scheme and unique identifier should be automatically usable by state-of-the-art smartphones and similar consumer devices, and compatible with all major operating systems and browsers.

5 ID schemes for products

5.1 ID scheme 5.1: Web enabled, structured path and query ID for products

5.1.1 Description

This ID scheme ensures globally unique identification based on ISO/IEC 15459-4 (individual products) and ISO/IEC 15459-6 (grouped products). It structures identifiers using ISO/IEC 15418 and encodes them in URLs or resolves them via HTTP as per ISO/IEC 18975. Users can choose between two web-enabled approaches: a structured path format or a query string format. The scheme enables direct web access to digital product passports and supports offline parsing for supply chain and point-of-sale processes. This ID scheme is based on issuing agencies (3.7) as per ISO/IEC 15459-2.

5.1.2 Requirements

5.1.2.1 Web enabled, structured path ID for products

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO/IEC 15459-4:2014 for individual products and/or ISO/IEC 15459-6:2014 for product groupings to ensure identifier uniqueness,

b) apply the GS1 Application Identifier (AI) format in accordance with ISO/IEC 15418:2016,

c) follow the structured path approach specified in ISO/IEC 18975:2024 for encoding and resolving identifiers over HTTP, and

d) use a registered numerical issuing agency according to ISO/IEC 15459-2:2015.

5.1.2.2 Web enabled, query string ID for products

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO/IEC 15459-4:2014 for individual products and/or ISO/IEC 15459-6:2014 for product groupings to ensure identifier uniqueness,

b) apply the ASC MH10 Data Identifiers (DI) format in accordance with ISO/IEC 15418:2016,

c) follow the query approach specified in ISO/IEC 18975:2024 for encoding and resolving identifiers over HTTP, and

d) use a registered alphanumerical issuing agency according to ISO/IEC 15459-2:2015.

5.2 ID scheme 5.2: Identification Link (IL)

5.2.1 Description

This ID scheme, based on IEC 61406-1 and IEC 61406-2, enables unique product identification using Identification Links (IL) in URL format. It allows economic operators to encode existing product IDs (model, batch, or item) directly into the IL, ensuring integration with their IT systems. The IL is based on a self-issuing system (3.18), removing external dependencies while supporting additional structuring via ANSI MH10.8.2 Data Identifiers specified in ISO/IEC 15418. This approach ensures compatibility with existing ID schemes and enables direct web access to digital product passports.

5.2.2 Requirements

5.2.2.1 Item level identification (only)

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to EN IEC 61406-1:2022, with optional use of [2], when a unique string of characters is sufficient, and no syntactical or semantic intelligence is required.

5.2.3 Guidelines

The economic operator should use:

a) [3], IL, where the string of characters of the IL should be short, e.g. to be encoded into a 96-bit RFID chip or to allow a compact 2D-code for small products; or where protection against the threat of counting up the product serial number shall be intercepted by masking model and serial number with a random alphanumerical string.

b) [2] for comprehensive product identification across model, batch, and item levels, ensuring semantic interoperability with existing ID schemes. It also can contain internal further information about the kind of identified object, for example product, person, location or document or additional data elements or classifications.

5.2.4 Model, batch and item level identification

To ensure compliance with this ID scheme, the economic operator shall:

a) use EN IEC 61406-1:2022 combined with EN IEC 61406-2:2024 when encoding standardized syntax and semantics is required. This includes ANSI MH10.8.2 Data Identifiers (DI) as specified in ISO/IEC 15418:2016to define model, batch, and item identification.

5.3 ID scheme 5.3: Decentralized identifiers (DID) for products

5.3.1 Description

This ID scheme is a self-issuing system that utilizes Decentralized Identifiers (DIDs) for self-generated, globally unique identification. DIDs support cryptographic authentication and privacy preserving information management. The scheme ensures interoperability with centralized, federated, and decentralized ID systems. Methods such as did:web, did:ethr, and did:ebsi provide flexible implementations, with did:web offering a lightweight, non-DLT solution and did:ebsi/did:ethr enabling dynamic updates. This approach enhances security, flexibility, and adaptability for digital identity management.

5.3.2 Requirements

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to Decentralized Identifiers (DIDs) v1.0:2022 to ensure proper DID syntax, core properties, serialized representations, and resolving DIDs to the resources they represent.

The economic operator should also use one of the following:

b) DID methods: did:web [4], did:ethr [5] or did:ebsi [6].

5.4 ID scheme 5.4: Product and group identification

5.4.1 Description

This ID scheme is suitable for limited capacity identifiers, namely RFID and QR-codes, that utilize resolvers to access DPP. The alternative of encoding a URL is also supported. The method ensures globally unique identification using ISO/IEC 15459-4 (individual products) and ISO/IEC 15459-6

(grouped products). It structures and encodes data for RFID applications based on ISO/IEC 15961, ISO/IEC 15962 and ISO/IEC 17360 and uses ISO/IEC 15434 for encoding data in QR-codes. The scheme standardizes application communication with data processors and operates independently of the ISO/IEC 18000 air interface. It supports offline processing of product details and follows ISO/IEC 15459-2 for identifier allocation and syntax. For additional security, the economic operator may optionally apply ISO/IEC 20248 for digital signatures.

5.4.2 Requirements

5.4.2.1 Product and group identification, RFID

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO/IEC 15459-4:2014 for individual products, and/or ISO/IEC 15459-6:2014 for groupings of products, to ensure the uniqueness of product identifiers,

b) follow ISO/IEC 15961-1:2021 or ISO/IEC 17360:2023 for semantic structuring and logical representation of data, and

c) use a registered issuing agency according to ISO/IEC 15459-2:2015.

5.4.2.2 Product and group identification, 2D-symbols

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO/IEC 15459-4:2014 for individual products, and/or ISO/IEC 15459-6:2014 for groupings of products, to ensure the uniqueness of product identifiers,

b) structure the data transfer syntax for high-capacity automatic data capture media according to ISO/IEC 15434:2019 to encode, for instance, URLs in QR-codes.

c) use a registered issuing agency according to ISO/IEC 15459-2:2015.

5.4.3 Guidelines

If instrumental, the economic operator may additionally use:

a) [7] for RFID data encoding and decoding,

b) [8] to encode HTTP URIs in RFID,

c) [9] for digital signatures.

5.5 ID Scheme 5.5: Digital Object Identifier (DOI) for products

5.5.1 Description

This ID scheme, based on ISO 26324, enables unique identification of products using Digital Object Identifiers (DOI) in URL format. It allows economic operators to encode existing product IDs (model, batch, or item) directly into the DOI. The DOI is based on a flexible system of either self-issuing or assignment by a service provider depending on the practices of the enabling service provider. This ID scheme is based on issuing agencies as per ISO 26324. It allows compatibility with existing ID schemes and enables direct web access to digital product passports.

5.5.2 Requirements

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO 26324:2025 for model, batch or item identification,

b) use the ISO/IEC 15459-2:2015 assigned code at the start of the DOI.

6 ID schemes for economic operators and facilities

6.1 ID scheme 6.1: Structured path identification for organizations

6.1.1 Description

This ID scheme is based on ISO/IEC 6523-1, providing a standardized framework for unique identification of economic operators and facilities. The scheme allows hierarchical assignment of identifiers, enabling multinational companies to allocate unique IDs to parent entities, subsidiaries, and operational divisions, as well as to facilities such as factories, warehouses, and offices. It facilitates access to organizational data via web browsers and supports offline parsing for system integration and backup processes. Adhering to ISO/IEC 6523, it ensures identifiers follow defined structure, format, and syntax, promoting global interoperability in supply chain and administrative workflows.

6.1.2 Requirements

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to ISO/IEC 6523-1:2023 to ensure unique economic operator identifiers and unique facility identifiers,

b) use a registered ID scheme according to ISO/IEC 6523-2:1998.

6.1.3 Guidelines

If instrumental, the economic operator may additionally use:

a) [8] for encoding and resolving identifiers over HTTP,

b) [3] and/or [2] for constructing identifiers as URLs.

6.2 ID scheme 6.2: Legal Entity Identifier (LEI)

6.2.1 Description

This ID scheme is based on the ISO 17442 series, defining the Legal Entity Identifier (LEI) as a standardized, globally unique identifier for legal entities engaged in financial activities. Each LEI consists of 20 alphanumeric characters, structured with a 4-character prefix, a 2-character reserved segment, and a 14-character unique identifier. Managed by Local Operating Units (LOUs), the centralized registration system prevents duplication and ensures global interoperability. The LEI integrates with financial reporting systems, adhering to ISO 17442 for cross-platform compatibility. Governance is provided by the Global Legal Entity Identifier Foundation (GLEIF), ensuring compliance, standardization, and maintenance

6.2.2 Requirements

To ensure compliance with this ID scheme, the economic operator shall:

a) assign their unique economic operator identifiers according to ISO 17442-1:2020.

6.2.3 Guidelines

If instrumental, the economic operator may additionally use:

a) [8] for encoding and resolving identifiers over HTTP,

b) [3] and/or [2] for constructing identifiers as URLs.

6.3 ID scheme 6.3: Decentralized identifiers (DID) for organizations

6.3.1 Description

This Decentralized Identity (DID) framework is a self-issuing system that provides a globally unique, self-sovereign identifier that is based on W3C Decentralized Identifiers (DIDs). Each DID includes a public DID document that defines its properties and contains a public key for verification. A DID becomes an Economic Operator Identifier when a Company Register issues a Verifiable Credential (VC) to it. This enables economic operators to sign and issue verifiable, tamper-proof digital product passports. The public key for verification can be found in the public DID document, ensuring transparency and trust. Decentralized Authentication and Access Control can be implemented via Role Verifiable Credentials issued to the economic operator DID. This allows third parties to authenticate themselves for access to non-public Digital Product Passport (DPP) data.

6.3.2 Requirements

To ensure compliance with this ID scheme, the economic operator shall:

a) adhere to Decentralized Identifiers (DIDs) v1.0:2022 for creation of self-sovereign identities including public-private key pairs,

b) follow Verifiable Credentials Data Model v1.1:2022 to associate verifiable credentials to the economic operator DID or the facility DID,

c) use Decentralized Identifier Resolution (DID Resolution) v0.3:2025 to find the DID document and the attached verifiable credentials.

6.3.3 Guidelines

If instrumental, the economic operator may additionally:

a) use trusted company registries, for example, national company registries or global company registries, such as the Global Legal Entity Identifier Foundation (GLEIF). To bind a DID to a company, GLEIF would issue a vLEI (Verifiable Legal Entity Identifier) [10] based on a national company register (see also ID Scheme 1) to the DID at hand.

b) associate other relevant Verifiable Credentials with the economic operator DID—such as sustainability disclosures, compliance reports, or ISO certifications—to create a chained provenance trail.

c) for facilities, trusted companies or registries, such as the US Facility Registry or a GS1 GLN, can issue identification credentials to the facility DID. Auditors and regulatory bodies can then issue audit reports as Verifiable Credentials to that facility DID, such as due diligence reports, compliance audits, and risk assessments.

6.4 ID scheme 6.4: Digital Object Identifiers (DOI) for organizations

6.4.1 Description

This ID scheme, based on ISO 26324, enables unique identification of organizations and facilities using the Digital Object Identifiers (DOI) in URL format. This ID schema is based on issuing agencies as per ISO 26324. It allows the use of existing identifiers or allocation of new identifiers depending on the practices of the issuing agency. It allows compatibility with existing ID schemes and enables direct web access to information about the identified object.

6.4.2 Requirements

To ensure compliance with this ID scheme, the registrant shall:

a) adhere to ISO 26324:2025 to ensure unique economic operator and facility identifiers were sufficient metadata is required by the issuing agency to ensure uniqueness of the organization or facility to which the identifier is assigned.

Annex A
(informative)

Guidance on selecting item-level identification for products

Prior to the market release of a product, a determination should be made whether to assign identification at the model, batch or item level, balancing different consideration of regulatory, technical, ecological, and economic factors. Opting for the appropriate level of object identification can lead to significant cost and effort reductions throughout the product's life cycle.

Default identification at the item level is advisable for products that meet any of the following criteria:

— they can be repaired and/or refurbished and/or overhauled by changing parts or components,

— they can be upgraded/enhanced by changing parts or components,

— its (embedded) software can be upgraded,

— regular maintenance and/or inspection or statutory audit is required by the economic operator, or by regulation,

— its wear level and/or operated hours and/or any other information about its use, will be recorded.

— they have to be identified uniquely on item level to fulfil their intended purpose (e.g. a Bluetooth module).

In scenarios not explicitly covered above, but where item-level identification emerges as beneficial during the product life cycle, economic operators are encouraged to default to this mode of identification. For instance, gathering data on the installed product base for strategic product planning purposes may warrant such an approach.

In exceptional circumstances, a product typically identified at the model level may require subsequent identification at the item level later in its life cycle. In such cases, economic operators needing item-level identification should generate and apply a unique product identifier according to the standard practices outlined herein. This approach maintains the linkage to the original economic operator's product information (typically associated with batch-level identification) while enabling the assignment of specific data to the individual item through the additional identifier.

This guidance aims to streamline the decision-making process regarding granularity levels of product identification, ensuring compliance with regulatory requirements while optimizing ecological and economical outcomes.

Annex B
(informative)

Overview of ID schemes for products

B.1 General information

See Table C.2 for general information.

Table B.1 — General information

ID scheme	5.1.2.1	5.1.2.2	5.2.2.1	5.2.2.2	5.3	5.4.1.1	5.4.1.2	5.5
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link	Identification link (structured ID link)	Decentralized identifer	Identification for products and product groups, RFID	Identification for products and product groups, 2D-symbols	Digital object identifier
Identifier standard	ISO/IEC 15459 + AI subset of ISO/IEC 15418	ISO/IEC 15459 + DI subset of ISO/IEC 15418	EN IEC 61406-1	EN IEC 61406-1 + EN IEC 61406-2	DIDs v1.0:2022	ISO/IEC 15459 with ISO or GS1 subsets	ISO/IEC 15459 with ISO or GS1 subsets	ISO 26324
Uniqueness	ISO/IEC 15459	ISO/IEC 15459	Web domain (RFC 3986)	Web domain (RFC 3986), optionally with ISO/IEC 15459, DID or other ID	DIDs v1.0:2022	ISO/IEC 15459	ISO/IEC 15459	ISO 26324 + ISO/IEC 15459
Technical standard for syntax	ISO/IEC 15459 + RFC 3986	ISO/IEC 15459 + RFC 3986	RFC 3986	RFC 3986	RFC 3986	RFC 3986	RFC 3986	ISO 26324 + ISO/IEC 15459
Application standard syntax	GS1 Digital Link (subset of ISO/IEC 18975)	AutoID-URL 1.3 (compatible to query string subset of ISO/IEC 18975)	EN IEC 61406-1	EN IEC 61406-1 and EN IEC 61406-2 (compatible to query string subset of ISO/IEC 18975)	DIDs v1.0:2022	ISO/IEC 15961 (all parts), ISO/IEC 15962, ISO/IEC 17360, ISO/IEC 20248	ISO/IEC 17360, ISO/IEC 20248, ISO/IEC 15434	ISO 26324
Technical standard for semantics	ISO/IEC 15459 + ISO/IEC 18975	ISO/IEC 15418 + ISO/IEC 18975	N/A, structure-only, no embedded semantics	ANSI MH 10.8.2 Data Identifier (subset of ISO/IEC 15418)	DIDs v1.0:2022	ISO/IEC 15459, ISO/IEC 15962, ISO/IEC 18975	ISO/IEC 15418, ISO/IEC 15424	ISO 26324

Application standard for semantics	AI subset of ISO/IEC 15418	DI subset of ISO/IEC 15418	N/A – no embedded semantics	EN IEC 61406-2	DIDs v1.0:2022	ISO/IEC 15961, ISO/IEC 17460, ISO/IEC 18975, ISO/IEC 20248	ISO/IEC 15418, ISO/IEC 15424	ISO 26324
Syntactic interoperability	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Semantic interoperability	Yes, parse GS1 Digital Link or structured path subset of ISO/IEC 18975	Yes, parse query parameters (RFC 3986) and semantics via ANSI MH 10.8.2	No semantics by default	Yes, interpret parameters per IEC 61406-2, ANSI MH 10.8.2	Yes, self-describing or aliasing via "alsoKnownAs"	Yes, structured encoding supports semantic resolution	Yes, structured encoding supports semantic resolution	Partial, requires resolution to access associated semantics
Decoding of level of uniqueness (model, batch, item) from encoded data	From encoded structure: model, batch, item	From query parameters and data identifiers	Parse URL parameters (RFC 3986) and exclude names starting with a dot.	Structured identifiers support decoding of level (model, batch, item)	On QR codes: decode QR code, get link, open browser and let Internet resolve it.	From tag content or system data model	From 2D symbol content	Resolution provides access to descriptive data that will specify level of uniqueness

B.2 Issuing and ownership

See Table B.2 for information on issuing and ownership.

Table B.2 — Issuing and ownership

ID scheme	**5.1.2.1**	**5.1.2.2**	**5.2.2.1**	**5.2.2.2**	**5.3**	**5.4.1.1**	**5.4.1.2**	**5.5**
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link	Identification link (structured ID link)	Decentralized identifer	Identification for products and product groups, RFID	Identification for products and product groups, 2D-symbols	Digital object identifier
Type of issuing	Issuing agency	Issuing agency	Self-issuing system	Self-issuing system	Self-issuing system	Issuing agency	Issuing agency	Issuing agency
Owner of ID	Economic operator	Economic operator	Economic operator	Economic operator	Economic operator	Economic operator	Economic operator	Economic operator

Translation mechanism	No need for translation mechanism for DPP access.	No need for translation mechanism for DPP access.	No translation needed. Existing IDs can be used directly.	No translation needed. Existing IDs can be used directly.	No translation needed. DIDs are self-describing; optional aliasing via “alsoKnownAs”	No translation needed. Auto-discriminating, open RFID standards	No translation needed. Auto-discriminating, open 2D standards	Resolver always required – some translation is inherent

B.3 General technical properties

See Table B.3 for information on general technical properties.

Table B.3 — General technical properties

ID scheme	5.1.2.1	5.1.2.2	5.2.2.1	5.2.2.2	5.3	5.4.1.1	5.4.1.2	5.5
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link	Identification link (structured ID link)	Decentralized identifer	Identification for products and product groups, RFID	Identification for products and product groups, 2D-symbols	Digital object identifier
Number of identifiers	Unlimited.	Unlimited.	Practically unlimited IDs (66^250), no third-party needed, no risk of exhaustion.	Practically unlimited IDs (66^250), no third-party needed, no risk of exhaustion.	Unlimited, no third-party required, no risk of identifier limits.	Unlimited.	Unlimited.	Unlimited.
Physical data carrier: 2D	Can be encoded in 2D symbols, including QR codes and DataMatrix.	Any AIDC media, technology independent.	Can be encoded in 2D symbols (with IEC 61406 conformant frame).	Can be encoded in 2D symbols (with IEC 61406 conformant frame).	Any data carrier / data carrier agnostic	Possible, but not typical. Focus is on RFID.	Yes, 2D symbols like DataMatrix or QR.	Yes, commonly used with QR codes.
Physical data carrier: RFID	Yes, can be encoded in RFID/NFC (e.g., EPC URIs).	Yes, can be encoded in RFID/NFC.	Yes, can be encoded in RFID/NFC.	Yes, can be encoded in RFID/NFC.	Any data carrier / data carrier agnostic	Yes, primary carrier for this scheme.	Not typically used.	RFID, NFC
Reliance of domain name	No, unique identifier is independently of domain name.	No, unique identifier is independently of domain name.	Yes, uses domain name as part of the identifier for global uniqueness.	Yes, uses domain name as root, though alternative methods possible.	No, identifier is self-resolving using DID method syntax.	For URI identifiers: No. For ISO/IEC 20248: Yes.	For URI identifiers: No. For ISO/IEC 20248: Yes.	Yes, relies on centralized DOI resolution via domain infrastructure.

B.4 Requirements

See Table B.4 for information on requirements.

Table B.4 — Requirements

ID scheme	5.1.2.1	5.1.2.2	5.2.2.1	5.2.2.2	5.3	5.4.1.1	5.4.1.2	5.5
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identification link	Identification link (structured ID link)	Decentralized identifer	Identification for products and product groups, RFID	Identification for products and product groups, 2D-symbols	Digital object identifier
4.1 Global uniqueness								
4.1.2 (1) No duplication	Ensured by the rules set in ISO/IEC 15459.	Ensured by the rules set in ISO/IEC 15459.	Ensured by the rules in IEC 61406-1.	Ensured by the rules in IEC 61406-1 and IEC 61406-2.	Ensured by W3C Decentralized Identifiers (DIDs) v1.0	Ensured by the rules set in ISO/IEC 15459.	Ensured by the rules set in ISO/IEC 15459.	Checked by issuing agency – each ID is registered with unique metadata.
4.1.2 (2) Distinct	Yes, globally unique and context independent per ISO/IEC 15459-3.	Yes, globally unique and context independent per ISO/IEC 15459-3.	Yes, the web domain remains the anchor for global uniqueness.	Yes, the web domain remains the anchor for global uniqueness.	Yes, globally unique.	Yes, globally unique per ISO/IEC 15459-3	Yes, globally unique per ISO/IEC 15459-3	Yes, globally unique per ISO/IEC 15459-3 and ISO 26324
4.1.2 (3) Non-coexistence	Prohibits reusing or reassigning identifiers.	Prohibits reusing or reassigning identifiers.	Prohibits reusing or reassigning identifiers.	Prohibits reusing or reassigning identifiers.	Ensured by cryptography and DID method rules – reassignment and reuse are prohibited.	Prohibits reusing or reassigning identifiers	Prohibits reusing or reassigning identifiers	Prohibits reusing or reassigning identifiers
4.2 Persistence								
4.2.2 (1) Consistency	ISO/IEC 15459-3	ISO/IEC 15459-3	Ensured by the rules in EN IEC 61406-1 and the EU DPP System Registry.	Ensured by the rules in EN IEC 61406-1 and EN IEC 61406-2 and the EU DPP System Registry.	Ensured by W3C DID v1.0 – persistence is technology-driven, not process-based.	ISO/IEC 15459-3	ISO/IEC 15459-3	Ensured by requiring descriptive metadata for each entity provided to issuing agency.

4.2.2 (2) Preservation	Supported by life cycle management rules per ISO/IEC 15459-3.	Supported by life cycle management rules per ISO/IEC 15459-3.	Ensured by life cycle management rules per EN IEC 61406-1 and the EU DPP System.	Ensured by the life cycle management rules per EN IEC 61406-1 and -2 and the EU DPP System.	Through decentralized structure and governance throughout the life cycle.	Supported by life cycle management rules per ISO/IEC 15459-3.	Supported by life cycle management rules per ISO/IEC 15459-3.	Issuing agency responsible for preservation of identifiers.
4.2.2 (3) Permanence	Per ISO/IEC 15459-3, does not relay on persistency of domain name.	Per ISO/IEC 15459-3, does not relay on persistency of domain name.	Ensured by the DPP system – web domains can be transferred or redirected.	Ensured by the DPP system – web domains can be transferred or redirected.	DIDs remain valid long-term; permanence depends on the chosen DID method.	Per ISO/IEC 15459-3, does not relay on persistency of domain name.	Per ISO/IEC 15459-3, does not relay on persistency of domain name.	Issuing agency ensures identifiers remain permanently linked to item metadata.
4.3 Syntax								
4.3.2 (1) Character set	Numerical	Alphanumerical	Alphanumerical	Alphanumerical	Alphanumerical	Alphanumerical	Alphanumerical	ISO 26324 defines characters that may be used
4.3.2 (2) Web	URI HTTP and URL per RFC 3986, plus a few additional rules set in ISO/IEC 18975.	URI HTTP and URL per RFC 3986, plus a few additional rules set in ISO/IEC 18975.	URL according to RFC 3986, plus few restrictions specified in EN IEC 61406-1.	URL according to RFC 3986, plus few restrictions specified in EN IEC 61406-1 and EN IEC 61406-2.	URI according to RFC 3986.	URI per RFC 3986	URI per RFC 3986, plus additional rules set in ISO/IEC 15434.	URL per IETF RFC 3986 mandatory
4.3.3 (1) Structure	Predefined structure usable across operators and service providers	Based on ISO/IEC 15459 (Data Identifiers)	Can optionally integrate ISO/IEC 15459 (DIs), GUIDs, DIDs, or internal IDs (e.g., product name + serial number)	Can optionally integrate ISO/IEC 15459 (DIs), GUIDs, DIDs, or internal IDs (e.g., product name + serial number)	"did:" prefix with method (e.g., web, ethr, ebsi) and method-specific ID	ISO/IEC 15459 ISO/IEC 15961 ISO/IEC 15962 ISO/IEC 17360	ISO/IEC 15434	ISO 26324 – issued prefix + constructed suffix (by operator or agency)

4.3.3 (2) Variable length	A fixed-length identifier with optional variable-length extensions within the same hierarchy.	Yes	Variable between 6 and 255 characters	Variable between 6 and 255 characters	Variable	Combination of fixed, structured, and variable elements	Combination of fixed, structured, and variable elements	Variable – can be fixed depending on issuing agency requirements
4.4 Semantics								
4.4.2 (1) Granularity	Model, batch, item via qualifier.	Model, batch, item via Data Identifier (ANSI MH 10.8.2).	Item via graphical marking and parameter names.	Model, batch, item via Data Identifier or graphical marking.	Model, batch, item as DID document attributes.	Model, batch, item via qualifier.	Model, batch, item via qualifier.	Model, batch, item.
4.4.2 (2) Granularity consistency	Supports multiple levels.	Supports multiple levels.	Item level only.	Supports model, batch, item.	Granularity set in DID document.	Supports multiple levels.	Supports multiple levels.	Fixed once registered.
4.4.2 (3) Change in granularity	Change in granularity requires new identifier.	Change in granularity requires new identifier.	Change in granularity requires new identifier.	Change in granularity requires new identifier.	Allowed – managed via DID document.	Change in granularity requires new identifier.	Change in granularity requires new identifier.	Change in granularity requires new identifier.
4.4.3 (1) Granularity level	Allowed	Flexible	Only item level	Allowed	All levels supported	Allowed	Allowed	Flexible
4.4.3 (2) Resolver	Browser-based, no special resolver	Browser-based, no special resolver	DNS-based resolution	DNS-based resolution	DID resolver + DNS (per W3C spec)	DNS, HTTP, or DID-based resolution	DNS, HTTP, or DID-based resolution	HTTP resolution per ISO 26324
4.4.3 (3) Hierarchy	Yes	Yes	No	Yes	Yes	Yes	Yes	Per issuing agency
4.5 Interoperability								
4.5.2 (1) Data carrier	Data carrier agnostic and web-enabled.	Data carrier agnostic and web-enabled.	Web-enabled data carriers.	Web-enabled data carriers.	Data carrier agnostic and web-enabled.	RFID	2D-symbols	Web-enabled data carriers.
4.5.2 (2) Digital copy	Ok	Ok	Ok	Ok	Ok	Ok	Ok	Ok

4.5.2 (3) Compatibility	Compatible with most external components used in supply chain.	Compatible with most external components used in supply chain.	Compatible with most external components used in supply chain.	Compatible with most external components used in supply chain.	Compatible with all web technologies and external components.	Compatible with most external components used in supply chain.	Compatible with most external components used in supply chain.	Can be used with web technologies and existing supply chain equipment as per ISO/IEC 15459
4.5.2 (4) Portability	Standardized, cross-system	Portable across platforms	Globally unique, cross-platform	Globally unique, cross-platform	Globally unique, platform-independent	Standardized, cross-system	Standardized, cross-system	Standardized, cross-system
4.5.2 (5) Existing ID	Ok	Ok	Ok	Ok	Ok	Ok	Ok	OK
4.6 Openness								
4.6.2 (1) Transparency	Public ISO/IEC + web standards	Public ISO/IEC + web standards	Public EN IEC + web standards	Public EN IEC + web standards	Public W3C	Public ISO/IEC standards	Public ISO/IEC standards	Public ISO + ISO/IEC standards
4.6.2 (2) End consumer usage	No additional software needed.	No additional software needed.	No additional software needed.	No additional software needed.	No additional software needed.	IDs may need to be parsed and resolved.	IDs may need to be parsed and resolved.	No additional software needed.
4.6.2 (3) No undue restrictions Also see B.6	No lock-in	No lock-in	No lock-in, only requires domain registration, which is transferable between providers	No lock-in, only requires domain registration, which is transferable between providers	No lock-in, DIDs are fully controlled by the owner and do not rely on any third party.	No lock-in	No lock-in	No lock-in, operator can switch issuing agencies or methods; existing IDs remain valid
4.6.3 (1) Smartphones	QR/NFC via URLs, smartphone compatible	QR/NFC via URLs, smartphone compatible	QR/NFC via URLs, smartphone compatible	QR/NFC via URLs, smartphone compatible	DID links via QR/ NFC, smartphone compatible	IDs/URLs readable in browsers	IDs/URLs readable in browsers	QR/NFC via URLs, browser-compatible

B.5 Security

See Table B.5 for information on security.

Table B.5 — Security

ID scheme	5.1.2.1	5.1.2.2	5.2.2.1	5.2.2.2	5.3	5.4.1.1	5.4.1.2	5.5
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identificati on link	Identificati on link (structure d ID link)	Decentrali zed identifer	Identificati on for products and product groups, RFID	Identificati on for products and product groups, 2D-symbols	Digital object identifier
Combine with ISO/IEC 20248	Yes, supports digital signature for data authentica tion.	Yes, supports digital signature for data authentica tion.	Yes	Yes		Yes, supports digital signature for data authentica tion.	Yes, supports digital signature for data authentica tion.	
Combine with ISO 22376	Yes, compatible for data element structure and transfer syntax.	Yes, compatible for data element structure and transfer syntax.	Yes	Yes		Yes, compatible for data element structure and transfer syntax.	Yes, compatible for data element structure and transfer syntax.	
Use of public/ private key pair	Yes, required for signing and verifying data when using ISO/IEC 20248.	Yes, required for signing and verifying data when using ISO/IEC 20248.	Not by default, but can be combined with	Not by default but can be combined with	Yes, DIDs come with security on board.	Yes, required for signing and verifying data when using ISO/IEC 20248.	Yes, required for signing and verifying data when using ISO/IEC 20248.	

B.6 Issuing process and prerequisites

See Table B.6 for information on issuing process and prerequisites.

Table B.6 — information on issuing process and prerequisites

ID scheme	5.1.2.1	5.1.2.2	5.2.2.1	5.2.2.2	5.3	5.4.1.1	5.4.1.2	5.5
Title	Web enabled structured path ID for products	Web enabled query string ID for products	Identificati on link	Identificati on link (structure d ID link)	Decentrali zed identifer	Identificati on for products and product groups, RFID	Identificati on for products and product groups, 2D-symbols	Digital object identifier

Prerequisite for web enabled access to DPP	Registered web domain of company or service provider (e.g. issuing agency)	Registered web domain of company or service provider (e.g. issuing agency)	Registered web domain of company or service provider	Registered web domain of company or service provider	Registered web domain of company or service provider	Not web-enabled	Not web-enabled	Registered web domain of company or service provider (e.g. issuing agency)
Process of issuing of product identifier	1. Get CIN from ISO/IEC 15459 agency 2. Issue product ID	1. Get CIN from ISO/IEC 15459 agency 2. Issue product ID	1.) Issue ID based on web domain	1.) Issue ID based on web domain	For did:web: 1.) Use web domain 2.) Create DID 3.) Publish DID document	1. Get CIN from ISO/IEC 15459 agency 2. Issue product ID	1. Get CIN from ISO/IEC 15459 agency 2. Issue product ID	Agency provides or approves prefix; ID issued by operator or agency
Sovereignty about the data in the product identifier	Operator controls ID elements.	Operator controls ID elements.	Operator controls self-issued elements	Operator controls self-issued elements	Full control; verifiable via cryptography	Operator controls product IDs	Operator controls product IDs	ID not reusable; operator holds admin control
Obligations of economic operator from technical terms in contracts	Contract with issuing agency	Contract with issuing agency	Only domain registrar contract, if no third-party used	Only domain registrar contract, if no third-party used.	No contract – self-issued	ISO/IEC 15459; optional X.509 cert (ISO/IEC 20248)	ISO/IEC 15459; optional X.509 cert (ISO/IEC 20248)	Contract with issuing agency.
Changes in technology that would effect the product identifier	Changes in ISO/IEC standards or RFC 3986.	Changes in ISO/IEC standards or RFC 3986.	Changes in ISO/IEC standards or RFC 3986.	Changes in ISO/IEC standards or RFC 3986.	Changes in W3C DID standards or RFC 3986.	Changes in ISO/IEC standards or RFC 3986.	Changes in ISO/IEC standards or RFC 3986.6.	Changes in ISO/IEC standards or RFC 3986.
Ceases of the service of a contracted service provider	Not applicable	Not applicable	Only domain registrar contract to manage.	Only domain registrar contract to manage.	No contracts – self-issued.	Not applicable.	Not applicable.	Fallback ensured by other agencies
Termination of contracts	Business decision	Business decision	Web domain contracts easily terminated	Web domain contracts easily terminated	Not applicable – self-managed; optional alias via "alsoKnownAs".	No restrictions.	No restrictions.	Set by issuing agency.

Conclusion of new contracts	Business decision	Business decision	Domain services can be contracted flexibly	Domain services can be contracted flexibly	Not applicable	No restrictions.	No restrictions.	Set by issuing agency
Unbundling of services	Yes	Yes	Domain transfer is standard and automated.	Domain transfer is standard and automated.	Not applicable	No restrictions.	No restrictions.	Ownership transfer supported by DOI agencies.
Procedures for switching a service provider	No restrictions.	No restrictions.	Domain transfer is standard and automated.	Domain transfer is standard and automated.	Not applicable.	No restrictions.	No restrictions.	Switching between agencies allowed.
Reference to an online register, where the product identifier or elements of it are registered	Not applicable	Not applicable	Domain Name registry	Domain Name registry	Not applicable	No restrictions.	No restrictions.	Maintained by issuing agency.
Additional services required to issue and maintain the product identifier	No additional services needed.	No additional services needed.	Only web domain required.	Only web domain required.	See resolver; examples may vary.	No additional services needed	No additional services needed	No additional services needed

B.7 Examples

B.7.1 ID scheme 5.1: Web enabled, structured path and query string identification for products

B.7.1.1 ID scheme 5.1.2.1: Web enabled, structured path identification for products

See Table B.7 for an example of web enabled structured path identification for products.

Table B.7 — Web enabled, structured path identification for products

Component	Meaning	Value
GTIN (01)	Global Trade Item Number	09524000059109
Serial Number (21)	Unique item-level identifier	1234
Date of Expiry (11)	YYMMDD format	250101

Component	Meaning	Value
Data carrier	QR-code example	
Scheme & Domain	https://example.com	
Full Example	https://example.com/01/09524000059109/21/1234?17=250101	

B.7.1.2 ID scheme 5.1.2.2: Web enabled, query string identification for products

See Table B.8 for an example of web enabled query string identification for products.

Table B.8 — Web enabled, query string identification for products

Component	Meaning	Value
Product ID (.25P)	IAC + CIN + Product Number	QCELMI12345
Serial number (.S)	Unique serial number	654321
Data carrier	QR-code example	
Scheme & Domain	HTTPS://WWW.DOMAIN-ABC.COM	
Full Example	HTTPS://WWW.DOMAIN-ABC.COM/?.25P=QCELMI12345&.S=654321	

B.7.2 ID scheme 5.2: Identification Link (IL)

B.7.2.1 ID scheme 5.2.2.1: Identification Link (IL)

See Table B.9 for an example of Identification Link (IL) without structure.

Table B.9 — Identification Link (IL) without structure

Component	Meaning	Value
Identifier (freetext)	Unstructured ID string	freetext
Data carrier	QR-code example with graphical frame to indicate item.	
Scheme & Domain	https://www.domain-abc.com	
Full example	https://www.domain-abc.com/freetext	

B.7.2.2 ID scheme 5.2.2.2: Identification Link (IL) with structure

See Table B.10 for an example of Identification Link (IL) with serial number per ISO/IEC 15418 used for item and batch.

Table B.10 — Identification Link with serial number per ISO/IEC 15418

Component	Meaning	Value
Scheme & Domain	Protocol and domain	https://www.domain-abc.com
Serial Number (.S)	Serial number, free format or per standard	freetext
Batch Number (.10B)	Batch or lot number	freetext
Data carrier	QR-code example with graphical frame to indicate batch or item.	
Full example:	https://www.domain-abc.com/?.S=freetext&.10B=freetext	

See Table B.11 for an example of Model Identification (Product Code Level with optional reuse of existing ID solutions).

Table B.11 — Model Identification (Product Code Level with optional reuse of existing ID solutions)

Component	Meaning	Value
Scheme & Domain	Protocol and domain	https://www.domain-abc.com
Product Code (.P)	Model-level identifier	freetext *(or from ID scheme)*
Data carrier	QR-code example with graphical frame to indicate model.	
Full example	https://www.domain-abc.com/?.P=freetext *(or other ID scheme)*	

B.7.3 ID scheme 5.3: Decentralized identifiers (DIDs) for products

See Table B.12 for an example of Decentralized Identifiers (DIDs) for products.

Table B.12 — Decentralized identifiers (DIDs) for products

Component	Meaning	Value
Product DID	Full DID identifying a product (DID method = web)	did:web:abc.com:model4TR
Service Endpoint	Optional service query parameter – specifies what to retrieve	?service=item-dpp
DID Method	Method part of the DID – determines resolution rules	web
Resolution Standard	Resolver must follow Decentralized Identifier Resolution (DID Resolution) v0.3:2025	W3C-compliant
Returned Resource	Based on service: may return a UI, machine-readable Verifiable Credential, or other digital product passport data	DPP (e.g. for item)

Component	Meaning	Value
Data carrier	QR-code example	
Scheme & Domain	https://resolver.io (DID resolver domain used to resolve the DID)	
Full example	https://resolver.io/did:web:abc.com:model4TR/?service=item-dpp	

B.7.4 ID scheme 5.4: Product and group identification

B.7.4.1 ID scheme 5.4.2.1: Product and group identification, RFID

See Table B.13 for an example of identification based on ISO-registered RAIN Alliance System.

Table B.13 — Identification based on ISO-registered RAIN Alliance System

Component	Meaning	Value
Protocol Control Word	Includes Numbering System Indicator (NSI)	31AE
Company ID	Company Identification Number (assigned under ISO by RAIN Alliance)	ABC001
Product Code (NDC)	National Drug Code – product identifier	01AB
Lot Number	Batch or production lot	02CD
Expiration Date	Coded expiration date (per issuer's internal rules)	04FC
Serial Number	Unique serial number per item	000001
Full Example (raw)	31AE ABC001 01AB 02CD 04FC 000001	
URN Representation (optional)	urn:rain:abc001:01AB:02CD:04FC:000001	

See Table B.14 for an example of identification based on SGTIN-96 (Pure Identity URI).

Table B.14 — Identification based on SGTIN-96 (Pure Identity URI)

Component	Meaning	Value
Company Prefix	GS1 Company Prefix	086699
Item Reference	GTIN Item Reference	0010034
Serial Number	Unique serial number for the item	10140309
Full example	urn:epc:tag:sgtin-96:0.086699.0010034.10140309	
NOTE Business-level ID (used in EPCIS and data exchange)		

See Table B.15 for an example of identification based on SGTIN-96 (Tag URI).

Table B.15 — Identification based on SGTIN-96 (Tag URI)

Component	Meaning	Value
Company Prefix	GS1 Company Prefix	086699
Item Reference	GTIN Item Reference	0010034
Serial Number	Unique serial number for the item	10140309
Full example	urn:epc:tag:sgtin-96:0.086699.0010034.10140309	

NOTE RFID-centric identifier (includes filter value)

B.7.4.2 ID scheme 5.4.1.2: Product and group identification, 2D-symbols

See Table B.16 for an example of product and group identification, 2D-symbols.

Table B.16 — Product and group identification, 2D-symbols

Component	Meaning	Value
GTIN (10)	Global Trade Item Number used as a base for batch-level identification (ISO/IEC 15459-6)	09506000134352
Batch Number (17)	Batch/lot identifier (e.g. production or packaging batch)	250101
Data Carrier Syntax	Encoded using high-capacity AIDC media syntax (e.g. QR code) per ISO/IEC 15434	Encoded in QR Code using ISO/IEC 15434
Resolution	Web resolver returns batch-level Digital Product Passport	Handled via example.com
Data carrier	QR-code example	
Scheme & Domain	https://example.com	
Full example	https://example.com/10/09506000134352/17/250101	

B.7.5 ID Scheme 5.5: Digital Object Identifier for products

See Table B.17 for an example of identification with DOI and ISO/IEC 15459 structure.

Table B.17 — Identification with DOI and ISO/IEC 15459 structure

Component	Meaning	Value
Scheme & Domain	DOI resolver domain	https://doi.org
Path Prefix (XID)	Identifier prefix signalling ISO/IEC 15459-compliant structure	XID
DOI Indicator	Constant prefix defined in ISO 26324 for DOI	10
Issued Org Code	Code assigned by the DOI issuing agency (registered under ISO/IEC 15459)	21
Product ID (Suffix)	Unique ID assigned by the org or agency (e.g. product, batch, item)	PRW82MJF
Resolution	DOI resolver translates to the Digital Product Passport location	Handled via doi.org
Data carrier	QR-code example	
Full Identifier	ISO 26324 compliant identifier including prefix and suffix	XID10.21/ PRW82MJF
Full example	https://doi.org/XID10.21/PRW82MJF	

Annex C
(informative)

Overview of ID schemes for economic operators and facilities

C.1 General information

See Table C.1 for general information.

Table C.1 — General information

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) fororganizations
Identifier standard	ISO/IEC 6523	ISO 17422	DIDs v1.0:2022	ISO 26324
Unique economic operator identifier	Yes	Yes	Yes	Yes
Unique facility identifier	Yes	No	Yes	Yes
Uniqueness	ISO/IEC 6523-1	ISO 17422-1	DIDs v1.0:2022	ISO 26324
Interoperability with other ID schemes.	Interoperability with all identifiers issued according to ISO/IEC 15459 by an issuing agency, as it follows the same structure and logic.	-	Yes. Can make different economic operator/ facility identifier schemes interoperable, such as ISO/IEC 15459 and GLEIF identifiers.	Other ID schemes can be used within DOIs as long as the uniqueness is maintained within the prefix allocated for use by economic operator

C.2 Issuing and ownership

See Table C.2 for information on issuing and ownership.

Table C.2 — Issuing and ownership

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) for organizations
Type of issuing	Issuing agency	Issuing agency	Self-issuing system	Issuing agency
Owner of ID	Economic operator	Economic operator	Economic operator	Economic operator
Translation mechanism	Not needed.	Not needed.	Not needed.	Not needed.

C.3 General technical properties

See Table C.3 for information on general technical properties.

Table C.3 — General technical properties

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) for organizations
Number of identifiers	Unlimited	Unlimited	Unlimited	Unlimited
Physical data carrier	Optional	Optional	Optional	Optional
Reliance of domain name	No	No	No	Yes
Web enabled	Yes, can be included in a URL; scannable by smartphones for direct access to product info.	Yes, can be included in a URL; scannable by smartphones for direct access to product info.	W3C standard; inherently web-enabled via resolvers and service endpoints.	Resolver ensures every DOI returns information about the referent.

C.4 Requirements

See Table C.4 for information on requirements.

Table C.4 — Requirements

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) for organizations
4.1 Global uniqueness				
4.1.2 (1) No duplication	Yes	Yes	Yes	Yes, registered with issuing agency and requires disambiguation metadata to prevent duplication.
4.1.2 (2) Distinct	Yes, ISO/IEC 6523-2	Yes, ISO 17422-1	Yes, DIDs v1.0:2022	Yes, issuing agency checks that identifier is unique within prefix
4.1.2 (3) Non-coexistence	Yes, prohibits reusing or reassigning identifiers.	Yes, prohibits reusing or reassigning identifiers.	Yes, prohibits reusing or reassigning identifiers (enforced by cryptography)	Yes, identifiers connected with referent through referent description – no reuse allowed
4.2 Persistence				
4.2.2 (1) Consistency	Yes	Yes	Yes	Yes, ensured by requiring descriptive metadata for each entity provided to issuing agency.
4.2.2 (2) Preservation	Yes	Yes	Yes	Yes
4.2.2 (3) Permanence	Yes, rules depend on issuing agency.	Yes, via GLEIF persistence.	Yes, under the control of the DID owner.	Yes, issuing agencies require the identifiers remain associated with the described referent
4.3 Syntax				

4.3.2 (1) Character set	Alphanumeric	Alphanumeric	Alphanumeric	ISO 26324 defines characters that may be used.
4.3.3 (1) Structure	Predefined (ICD + Org ID).	Fixed structure.	Partly fixed (did:<method>:).	Prefix/Suffix structure.
4.3.3 (2) Variable length	Variable within limits	Fixed length (20 chars)	Variable	Variable suffix
4.4 Semantics				
4.4.3 (2) Resolver	Optional	Yes (via GLEIF API)	Yes (DID resolver)	Yes (via doi.org)
4.4.3 (3) Hierarchy	Depending on ID scheme	No	Yes	No
4.5 Interoperability				
4.5.2 (1) Data carrier	Compatible with 2D/ RFID	Encodable in QR/ RFID	Yes any, if encoded	Encodable in QR/ RFID
4.5.2 (2) Digital copy	Yes	Yes	Yes	Yes
4.5.2 (3) Compatibility	Yes, with external components used in supply chain.	Yes, with external components used in supply chain.	Yes	Yes, can be used with web technologies
4.5.2 (4) Portability	Yes	Yes	Yes	Yes
4.5.2 (5) Existing ID	-	-	Semantics are added using verifiable credentials.	Specified in ISO 26324
4.6 Openness				
4.6.2 (1) Transparency	Open, ISO standard	Public, managed by GLEIF	Open W3C standard	Open, ISO-based
4.6.2 (2) End consumer usage	Yes, no additional software needed.	Yes, no additional software needed.	Yes, verifiability of identifier data.	Yes, no additional software needed. Can be resolved to access any publicly available referent information.
4.6.2 (3) No undue restrictions Also see C.6	No lock-in	No vendor lock-in	No lock-in, self-sovereign	No undue restrictions
4.6.3 (1) Smartphones	QR codes, RAIN tags, NFC enabled via ISO/IEC 18975. Compatible with smartphones, major operating systems, and browser.	QR codes, RAIN tags, NFC enabled via ISO/IEC 18975. Compatible with smartphones, major operating systems, and browser.	Hardware agnostic. Compatible with any device that runs a web browser.	Compatible with smartphones, major operating systems, and browser.

C.5 Security

See Table C.5 for information on security.

Table C.5 — information on security

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) for organizations

Combine with ISO/IEC 20248	Possible, not standard practice	Not typical	Yes, for verifiable credentials and metadata signatures	Possible, especially for content authentication
Combine with ISO 22376	Yes, for traceability and provenance documentation	Possible if mapped to products	Yes, well-aligned with credential-based traceability	Yes, especially with XID-structured DOIs
Use of public/private key pair	No (unless paired with 20248 or other mechanisms)	No (central registry model)	Yes, integral to DID authentication and verification	Optional – depends on resolution and registry design

C.6 Issuing process and prerequisites

See Table C.6 for information on issuing process and prerequisites.

Table C.6 — information on issuing process and prerequisites

ID scheme	6.1	6.2	6.3	6.4
Title	Structured path identification for organizations	Legal Entity Identifer (LEI)	Digital Object Identifiers (DOI) for organizations	Digital Object Identifiers (DOIs) for organizations
Prerequisite for web enabled access to DPP	Can be embedded in URL	Resolvable via GLEIF services	DID resolver and method required	Resolvable via DOI resolver
Process of issuing of product identifier	Get ICD and Org ID, then issue	Register via LOUs under GLEIF	Self-issued using DID method	Apply via DOI agency
Sovereignty about the data in the product identifier	Controlled by the economic operator	Maintained by economic operator and LOU	Full sovereignty by economic operator	Shared between economic operator and agency
Obligations of economic operator from technical terms in contracts	Contract with issuing authority	Based on GLEIF and LOU rules	None – self-governed	Depends on DOI agency
Changes in technology that would affect the product identifier	Minimal impact	Low impact, long-term stable	Depends on DID method	Resolver dependency
Ceases of the service of a contracted service provider	Not applicable	Other LOUs take over	Not applicable	Fallback via DOI federation
Termination of contracts	Yes	Standard process exists	Not applicable	Yes
Conclusion of new contracts	Yes	Yes	Freely possible	Yes
Unbundling of services	Yes	Yes	Yes	Yes
Procedures for switching a service provider	Yes	Yes	Yes	Yes
Reference to an online register, where the product identifier or elements of it are registered	ICD and Org ID in ISO registry	GLEIF public register	Public DID document	DOI agency registry

Additional services required to issue and maintain the product identifier	None	None	No, unless using certs	Resolver service needed

C.7 Examples

C.7.1 ID scheme 6.1: Structured path identification for organizations

See Table C.7 for a few examples of structured path identification for organizations, ID schemes included in ISO/IEC 6523-1.

Table C.7 — Structured path identification for organizations

Name of ID scheme	ICD Code	Description
GS1 Global Location Number (GLN)	0088	Used to identify legal entities, functional entities, and physical locations globally.
Dun & Bradstreet (DUNS)	0060	Unique business identifiers used globally for legal entities and company verification.
SIRENE (France)	0012	Used to identify French companies and establishments.
VAT Identifier (Germany)	0195	Identifier for German companies (based on VAT registration).
Legal Entity Identifier (LEI	0199	Identifies legal entities involved in financial transactions. Used for regulatory reporting, finance, and in supply chain transparency.

C.7.2 ID scheme 6.2: Legal Entity Identifier (LEI)

See Table C.8 for an example of a Legal Entity Identifier (LEI).

Table C.8 — Legal Entity Identifier (LEI)

Component	Meaning	Value
LEI	20-character LEI, issued under ISO 17442.	5493001KJTIIGC8Y1R12
Scheme & Domain	https://www.domain-abc.com	
Example 1: LEI (020)	LEI in ISO/IEC 18975 format	5493001KJTIIGC8Y1R12
Full example 1	https://www.domain-abc.com/020/5493001KJTIIGC8Y1R12	
Example 2: LEI (.020)	LEI in EN IEC 61406-2 format	5493001KJTIIGC8Y1R12
Full example 2	https://www.domain-abc.com/?.020=5493001KJTIIGC8Y1R12	

C.7.3 ID Scheme 6.3: Decentralized identifiers (DIDs) for organizations

See Table C.9 for an example of Decentralized identifiers (DIDs) for organizations.

Table C.9 — Decentralized identifiers (DIDs) for organizations

Component	Meaning	Value
DID Scheme	DID method prefix as per DIDs v1.0:2022	did:
Method	DID method (e.g., web, ethr, ebsi)	web
Domain & Path	Fully qualified domain controlled by the economic operator	abc.com:facility:plant-12
Metadata	DID Document may include public keys, service endpoints, facility roles, granularity, etc.	Stored in DID Document
Full example	did:web:abc.com:facility:plant-12	

C.7.4 ID Scheme 6.4: Digital Object Identifiers (DOIs) for organizations

See Table C.10 for an example of Digital Object Identifiers (DOIs) for organizations.

Table C.10 — Digital Object Identifiers (DOIs) for organizations

Component	Meaning	Value
Scheme & Domain	Protocol and resolver domain	https://doi.org
Prefix (10.23)	Issuing agency prefix assigned under ISO 26324	10.23
Suffix	Identifier created by the economic operator or issuing agency	BB55-375D-4B8F-A0DD-3FEB
Full DOI	Complete identifier (prefix + suffix)	10.23/BB55-375D-4B8F-A0DD-3FEB
Resolution	Handled by DOI resolver, redirects to Digital Product Passport (DPP) URL	Redirect managed by doi.org
Full example	https://doi.org/10.23/BB55-375D-4B8F-A0DD-3FEB	

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of 2024/1781

This European Standard has been prepared under a Commission's standardization request M/604to provide one voluntary means of conforming to essential requirements of Article 10 and 11 of 2024/1781 Ecodesign for Sustainable Product Regulation (ESPR).

Once this standard is cited in the Official Journal of the European Union under that 2024/1781, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding requirements of that 2024/1781, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and the essential requirements of Article 10 and 11 of 2024/1781

Essential requirements specified in Article 10 and 11 of 2024/1781	Sub-clause(s) of this EN	Remarks/Notes
Article 10, (1), (a)	4.1.2, 4.2.2, 4.5.2	Ensures the identifier is unique, persistent, and encoded in a data carrier.
Article 10, (1), (b)	N/A	-
Article 10, (1), (c)	N/A	-
Article 10, (1), (d)	4.3.2, 4.4.2, 4.5.2, 4.6.2	Ensures structured, machine-readable data using open, interoperable standards without vendor lock-in.
Article 10, (1), (e)	4.6.2	Public DPP information is accessible without credentials, preventing unauthorized personal data storage.
Article 10, (1), (f)	4.4.2	Supports identifier granularity (model, batch, item) and traceability through consistent linking.
Article 10, (1), (g)	N/A	-
Article 10, (2)	N/A	-
Article 10, (3), (a)	4.5.2	Retailers and marketplaces can access and use identifiers where physical access is not possible.
Article 10, (3), (b)	N/A	-
Article 10, (4)	N/A	-
Article 11, (a)	4.5.2, 4.6.2	Open standards and no vendor lock-in enable interoperability across DPP systems.
Article 11, (b)	4.6.2	Open, non-discriminatory access ensures availability of public DPP data to all stakeholders.
Article 11, (c)	4.2.2	Ensures DPP persistence even after insolvency or cessation, supported by registry use.
Article 11, (d)	4.4.2	New DPPs are linked to originals for traceability when changes in granularity occur.
Article 11, (e)	4.2.2	DPP remains available throughout life cycle, including after economic operators exit.
Article 11, (f)	N/A	-

Essential requirements specified in Article 10 and 11 of 2024/1781	Sub-clause(s) of this EN	Remarks/Notes
Article 11, (g)	4.6.2	Open and transparent design supports data authenticity, reliability, and integrity.
Article 11, (h)	4.6.2	Transparency and minimal access barriers help ensure privacy, security, and fraud prevention.
Article 11 (2)	N/A	-

WARNING Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING Other Union legislation may be applicable to the products falling within the scope of this standard.

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